

ANDRES SALAZAR

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SUMMARY

Mechanical Engineer with experience in aerodynamics, thermodynamics, and mechanical design, including FEA simulations, GD&T, ISO drawings and optimization for manufacturability. Skilled in fatigue, thermal evaluation and prototyping, with full-time availability starting January 2026.

EDUCATION

University of Houston

Bachelor of Science in Mechanical Engineering

Minor in Mathematics

Houston, TX

December 2025

- *Relevant Coursework:* Computer Design of Structures and Materials, Fluid Mechanics, Design of Machine Elements, Finite Elements, Design of Control Systems, Heat Transfer, Thermal Design, Aerodynamics, Conduction and radiation.

Cumulative College GPA: 3.62

SKILLS

Certifications: SolidWorks Professional Design, Fusion360, Six Sigma Green Belt, Excel – Copilot Professional

Software: CAM, COMSOL, ANSYS, MATLAB, Arduino, Python, UltiMaker, Excell, PowerPoint, Teams and Outlook.

Languages: Spanish (fluent) and English (fluent)

WORK EXPERIENCE

Engineering Practicum – University of Houston

Teaching Assistant – MECE Department

Houston, TX

May 2024 – December 2025

- Spearhead the development of a hands-on practicum course for 200+ students, equipping freshman and sophomore with industry-standard skills in CAD, 3D Printing, CNC Milling, and Laser Cutting while following safety protocols.
- Design structured, real-world-driven curriculum, creating 50+ CAD models, detailed lectures, quizzes, and assignments to enhance engagement and technical proficiency.

Workshop Facilitator – University of Houston

Teacher Assistant – MECE Department

Houston, TX

January 2024 – May 2024

- Mentored practical preparation for SolidWorks certification, resulting in 100+ students earning CSWA credentials and raising the certification rate from under 30% to over 80%.

Unmanned Aerosol System – Snow College

Research Assistant – Engineering Department

Ephraim, UT

October 2022 – March 2023

- Built a UAS platform to profile vertical temperature, relative humidity, pressure, and particulates, integrating an iMET XQ-2 reference with low-cost Arduino/BME280 and SPS30 sensors.
- Validated sensors via Part-107 flights, generating profiles to ~140 m (400 ft) AGL; confirmed iMET XQ-2's faster temperature response, while assessing durability and accuracy limitations of the BME280.
- Presented findings to the Utah Department of Environmental Quality.

RELEVANT PROJECTS

Lunar Dust Tolerant Electrical Connector – NASA JSC Power and Propulsion Division

University of Houston | Capstone Design Project |

Houston, TX

January 2025-December 2025

- Designed and prototyped a dust-sealed electrical connector for lunar EVA applications, ensuring IP6X-rated protection against abrasive lunar regolith with an innovative gate-sealing mechanism.
- Engineered a one-handed, EVA-glove-compatible operation with integration with robotic systems.
- Conducted FEA (ANSYS) and thermal simulations, including fatigue and vibration analysis, to validate structural integrity and material expansion from -170°C to +127°C in a vacuum environment.
- Collaborated with NASA mentors and suppliers to resolve manufacturability and tolerance challenges, ensuring readiness for prototype fabrication.

Activities

Machinist Assistant (Manufacturing Operations at UH)

August 2025 to December 2025

- Support CNC mill/lathe setups, work holding and tool offsets; perform G-code troubleshooting and shop safety.

Member of SHPE (Society of Hispanic Engineers)

August 2023 to December 2025

- Support professional development and outreach events that elevate Hispanic leadership in STEM.
- Build a mentor network through employer talks, resume reviews, and industry visits, gaining exposure to careers.

Member of ASME (Association of Mechanical Engineers)

August 2023 to December 2025

- Participate in technical talks, standards workshops, 3D print labs, and design/networking events.

Honors & Awards

- American Bureau of Shipping academic excellence scholarship
- Shell Engineering Award
- Degree Honors Magna Cum Laude

October 2025

February 2024

December 2025